

Exercise 4: Implementing the Adapter Pattern

CODE:

// Target Interface

```
interface PaymentProcessor {  
    void processPayment(double amount);  
}
```

// Adaptee 1

```
class PayPalGateway {  
    void makePayment(double amount) {  
        System.out.println("Payment of Rs." + amount + " processed using PayPal.");  
    }  
}
```

// Adaptee 2

```
class StripeGateway {  
    void payAmount(double amount) {  
        System.out.println("Payment of Rs." + amount + " processed using Stripe.");  
    }  
}
```

// Adapter for PayPal

```
class PayPalAdapter implements PaymentProcessor {  
    private PayPalGateway paypal;  
    PayPalAdapter(PayPalGateway paypal) {  
        this.paypal = paypal;  
    }  
    public void processPayment(double amount) {  
        paypal.makePayment(amount);  
    }  
}
```

// Adapter for Stripe

```
class StripeAdapter implements PaymentProcessor {  
    private StripeGateway stripe;  
    StripeAdapter(StripeGateway stripe) {  
        this.stripe = stripe;  
    }  
    public void processPayment(double amount) {  
        stripe.payAmount(amount);  
    }  
}  
  
// Main Class  
public class Main {  
    public static void main(String[] args) {  
        PaymentProcessor payment1 = new PayPalAdapter(new PayPalGateway());  
        PaymentProcessor payment2 = new StripeAdapter(new StripeGateway());  
        payment1.processPayment(2500);  
        payment2.processPayment(4800);  
    }  
}
```

OUTPUT:

```
Payment of Rs.2500.0 processed using PayPal.  
Payment of Rs.4800.0 processed using Stripe.
```